

LEEDALE AGDM, AB, Canada

WMO: 715650

Lat: 52.5529N

Lon: 114.4725W

Elev: 3107

StdP: 13.12

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.2	-19.2	-32.4	1.0	-26.1	-25.0	1.5	-18.9	25.0	26.6	22.2	20.6	2.4	160	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	80.5	62.6	77.4	61.2	74.5	59.9	65.8	76.5	63.8	74.5	61.7	71.6	7.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	92.6	72.2	59.5	85.3	69.0	57.5	79.3	66.6	32.4	76.5	30.7	74.7	29.1	71.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.2	16.6	-34.6	86.6	5.0	3.2	-38.2	88.9	-41.1	90.8	-43.9	92.6	-47.5	94.9
			-34.4	70.1	4.9	2.9	-38.0	72.2	-40.9	73.8	-43.7	75.5	-47.2	77.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	14.0	12.1	23.5	37.1	48.4	55.1	59.5	56.9	48.6	38.1	22.6	11.0	
	DBStd	20.19	13.78	14.15	12.93	8.85	6.95	5.02	4.94	5.27	7.14	7.80	13.09	13.35	
	HDD50	6038	1117	1061	820	393	114	11	1	7	109	376	821	1208	
	HDD65	10706	1582	1481	1285	838	515	299	182	253	492	835	1271	1673	
	CDD50	823	0	0	0	5	64	164	295	222	67	6	0	0	
	CDD65	16	0	0	0	0	0	2	11	3	0	0	0	0	
	CDH74	738	0	0	0	2	68	81	304	204	74	5	0	0	
	CDH80	106	0	0	0	0	7	7	54	31	7	0	0	0	
Wind	WSAvg	6.7	6.6	6.5	7.1	8.3	8.3	7.5	6.3	5.6	5.8	6.5	6.3	6.1	
Precipitation	PrecAvg	21.2	0.9	0.7	0.9	1.1	2.6	4.0	4.0	2.9	1.7	0.9	0.9	0.7	
	PrecMax	26.3	2.4	1.5	1.9	3.1	4.3	7.2	7.6	4.6	3.6	2.5	2.6	1.7	
	PrecMin	16.2	0.1	0.1	0.1	0.2	1.0	0.9	1.2	0.9	0.2	0.3	0.3	0.2	
	PrecStd	2.9	0.5	0.4	0.4	0.6	0.9	1.4	1.4	1.0	0.9	0.6	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	45.6	52.7	71.7	80.3	80.0	85.2	82.8	80.9	70.2	57.1	44.9	
		MCWB	37.7	37.1	41.4	50.9	55.6	62.4	66.2	63.1	59.9	53.1	44.3	35.9	
	2%	DB	42.9	39.6	47.2	65.1	75.3	76.4	80.9	79.2	75.5	62.8	49.3	38.5	
		MCWB	35.8	33.5	38.6	47.6	54.0	59.8	65.5	63.2	58.2	48.9	40.2	32.7	
	5%	DB	38.1	36.0	43.7	58.9	70.9	72.5	77.2	75.7	70.9	58.0	42.7	34.0	
		MCWB	33.0	31.4	36.5	43.7	52.0	57.9	63.8	62.0	56.4	46.2	36.2	29.8	
	10%	DB	33.3	32.2	40.0	53.3	66.1	68.9	73.9	71.9	65.3	52.8	38.4	30.1	
		MCWB	29.7	28.9	34.3	41.4	50.0	56.0	61.9	60.6	53.9	43.6	33.9	27.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	37.7	42.8	51.1	57.5	65.1	70.2	67.2	62.0	54.1	45.5	36.9
			MCDB	45.8	45.2	51.0	70.3	76.5	76.7	80.1	76.9	76.3	68.9	55.8	43.2
2%		WB	36.0	34.2	39.4	48.1	54.9	61.7	67.0	65.0	59.2	49.7	40.6	33.1	
		MCDB	42.4	38.9	46.6	64.1	71.3	72.3	77.7	75.8	72.4	61.3	47.7	37.5	
5%		WB	33.4	31.6	36.8	44.8	53.0	59.4	64.8	63.0	56.9	46.9	37.2	30.1	
		MCDB	38.0	35.4	42.8	56.9	68.0	69.6	75.5	73.6	69.1	56.1	41.8	34.0	
10%		WB	29.9	29.1	34.6	42.1	51.1	57.4	62.7	61.1	54.4	44.3	34.1	27.2	
		MCDB	33.5	32.3	39.3	51.8	64.4	67.0	72.4	71.0	64.6	51.8	37.6	30.2	
Mean Daily Temperature Range	5% DB	MDBR	21.8	23.4	21.7	22.7	26.2	22.9	24.5	25.7	25.9	24.5	20.1	20.5	
		MCDBR	26.0	25.9	25.2	35.2	37.2	30.4	31.1	34.1	38.6	33.5	25.5	23.9	
		MCWBR	19.6	20.4	17.5	19.0	17.9	15.8	17.3	19.6	21.7	19.9	17.2	18.6	
	5% WB	MCDBR	25.1	24.7	23.5	32.5	32.6	26.8	29.1	31.0	34.8	31.2	24.8	24.0	
		MCWBR	19.0	19.6	16.7	18.1	16.4	14.9	17.0	18.5	19.8	19.1	17.5	19.2	
Clear-Sky Solar Irradiance	taub		0.254	0.259	0.299	0.328	0.342	0.349	0.348	0.357	0.311	0.281	0.252	0.241	
	taud		2.344	2.414	2.393	2.374	2.385	2.387	2.427	2.371	2.483	2.558	2.489	2.343	
	Ebn at Noon		242	279	285	288	288	286	285	275	278	268	246	226	
	Edh at Noon		18	23	29	34	36	36	34	34	27	20	16	15	
All-Sky Solar Radiation	RadAvg		318	612	1055	1455	1719	1823	1934	1610	1128	677	356	244	
	RadStd		19	40	62	117	119	136	119	105	121	71	29	17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page